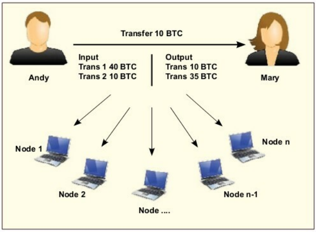

Figure 1: A bitcoin transaction between Andy and Mary

the amount of currency available to you for conducting transactions is the cumulative sum of all the transactions you've performed till date via that account. Now, as long as you don't withdraw the amount physically, it's just a number stored in the bank's database. When you perform an online transaction, using this account, it is recorded in your account —the relevant number is either deducted from your account or credited to it, depending on the nature of your transaction. Again, as long as you don't withdraw the balance amount physically, it exists in a virtual state in the bank's database. This is what bitcoin is, except that with bitcoins, you are your own bank; your bitcoin wallet is your bank account and you can't withdraw it physically. You can exchange it in return for some other currency but you can't have bitcoins in physical form because they only exist in digital (or virtual) form.

"Thou shalt remain in states of 0s and 1s forever…"

"Hey! Hold on…I have to first deposit 10,000 units of hard cash with the bank before they can become virtual and here you're saying that bitcoin only exists in virtual from. How does this work? Where are all these bitcoins coming from?"

Bitcoins originate via a process called bitcoin mining. Before I delve more into it, let's understand how the Bitcoin network works and how a transaction takes place. As I mentioned earlier, Bitcoin is a peer-to-peer network, which means that there's no single entity controlling it. It is a decentralised network and each system in this network acts as a node.

Let's assume that Andy and Mary are conducting a transaction in bitcoin (BTC), in which Andy wants to transfer 10 BTC to Mary in return for a service and he currently has a total of 50 BTC in his bitcoin wallet. Figure 1 depicts how the transaction between them would look.

There are two parts to this transaction—the input and output. The input part contains a reference to the previous transactions, which would account for the amount of BTC currently in Andy's wallet. The output part will contain the information of how many bitcoins Andy wants to transfer (10 BTC in this case) and how much change he wants back (35 BTC in this case). The difference between the amount he wants to transfer and the change he wants back, if any, is the transaction fee (5 BTC in this case). The transaction fee is the incentive for bitcoin miners to include that transaction in a transaction block (explained later). Andy will then sign this transaction with his digital signature and publish it to the Bitcoin network. All nodes in the Bitcoin network will receive this transaction. They may or may not act upon it, depending on their role in the network.

A bitcoin miner is a node in the Bitcoin network, running bitcoin mining software. A bitcoin mining node collects Andy's transaction, along with many others, and collates them into what is called a transaction block. If this block is accepted by the network, it will then be appended to the universal transaction block chain. Think of the transaction block chain as a universal ledger which has all the transactions recorded in it, since the beginning of the Bitcoin system. Once the miners' transaction blocks are accepted, they are rewarded with a certain number of bitcoins (in addition to the transaction fee) that accrues from all the transactions included in that block. This reward part is how new bitcoins are created in the Bitcoin network.

On the face of it, the bitcoin mining process might seem to be a very simplistic process, but it actually involves its share of complexity. The Bitcoin network doesn't simply accept a transaction block—a miner has to prove the effort he has put in while working on that transaction block. This is called 'proof of work'.

Proof of work is a cryptographic puzzle, which the Bitcoin network requires the miner to solve before the transaction block is accepted within the transaction block chain. The creation of a transaction block is fairly simple and doesn't require much effort; solving this puzzle is what makes mining a processing power-intensive task. The difficulty level of the puzzle is decided by the Bitcoin network, which is designed to adjust it, depending on how fast transaction blocks are being generated. If it senses that blocks are being generated too fast, the network increases the difficulty level and vice versa. The miner who solves the proof of work puzzle first, publishes the solution on the network. The Bitcoin network then verifies the solution sent by the miner, and if correct, the transaction block is appended to the transaction block chain. Though generating the solution is processing power-intensive, verifying it hardly takes any time.

While mining is one way in which bitcoins circulate, there are three much simpler ways:

- Exchanging bitcoins in return for some other currency, either from a designated exchange or from a person willing to make such an exchange. It's 100 per cent legal, as of date, though the Reserve Bank of India advises against it (*http://goo.gl/fbh3wO*).
- Accepting bitcoins for a service or product.
- Asking a friend for some (this might be bit tricky, given the current value of the bitcoin).